Business Finance Midterm

Multiple Choice · Chapters 2–7 · 70 questions

Exhibit A · Harborview Outfitters, Inc.

Used in questions 1–7

Balance sheets, December 31 ($ millions)

Item	2024	2025
Assets		
Cash	150	200
Accounts receivable	600	700
Inventory	850	900
Total current assets	1,600	1,800
Net fixed assets	4,400	4,600
Total assets	6,000	6,400
Liabilities and owners' equity		
Accounts payable	520	560
Notes payable	280	240
Total current liabilities	800	800
Long-term debt	1,800	1,900
Common stock and paid-in surplus	1,400	1,500
Retained earnings	2,000	2,200
Total owners' equity	3,400	3,700
Total liabilities and owners' equity	6,000	6,400

2025 income statement ($ millions)

Item	2025
Net sales	4,800
Cost of goods sold	2,900
Depreciation	500
Earnings before interest and taxes	1,400
Interest paid	200
Taxable income	1,200
Taxes (21%)	252
Net income	948
Dividends	748
Addition to retained earnings	200

1. [Exhibit A] Harborview Outfitters, Inc. ($ millions). The firm has 100 million shares outstanding and the share price is $74. What are its P/E, price-sales and market-to-book ratios?
   a. P/E 9.48; P/S 1.54; M/B 2.00
   b. P/E 7.81; P/S 1.54; M/B 2.00
   c. P/E 7.81; P/S 0.65; M/B 0.50
   d. P/E 7.81; P/S 1.54; M/B 0.50

2. [Exhibit A] Harborview Outfitters, Inc. ($ millions). What are the firm's 2025 times interest earned and cash coverage ratios?
   a. TIE 9.50; cash coverage 7.00
   b. TIE 7.00; cash coverage 9.50
   c. TIE 7.00; cash coverage 4.50
   d. TIE 6.00; cash coverage 8.50

3. [Exhibit A] Harborview Outfitters, Inc. ($ millions). On the 2025 common-size statements, what percentages are shown for inventory and for net income?

a. Inventory 18.75%; net income 14.81%

b. Inventory 14.06%; net income 19.75%

c. Inventory 50.00%; net income 25.62%

d. Inventory 14.06%; net income 14.81%

4. [Exhibit A] Harborview Outfitters, Inc. ($ millions). The firm has 100 million shares at $74 each. Enterprise value uses the book value of all liabilities. What is its EV/EBITDA ratio?

a. 5.21

b. 5.32

c. 3.89

d. 7.07

5. [Exhibit A] Harborview Outfitters, Inc. ($ millions). The firm has 100 million shares outstanding. Comparable outdoor-retail firms trade at 12 times earnings. What is the estimated share price?

a. $113.76

b. $88.80

c. $57.60

d. $9.48

6. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Using 2025 year-end balances, what are inventory turnover, receivables turnover and total asset turnover?

a. 3.22; 6.86; 0.75

b. 5.33; 6.86; 0.75

c. 3.22; 4.14; 1.33

d. 3.22; 6.86; 1.33

7. [Exhibit A] Harborview Outfitters, Inc. ($ millions). During 2025, which net effect on cash comes from the changes in accounts receivable and accounts payable alone?

a. $140 source of cash

b. $140 use of cash

c. $60 source of cash

d. $60 use of cash

8. Which statement about a bond that is already outstanding is correct?

a. The coupon rate is reset whenever market rates change

b. The coupon rate is fixed at issue, while the yield to maturity changes with the market every day

c. The yield to maturity is fixed at issue, while the coupon rate moves with the market

d. The face value rises when market rates fall

9. On a balance sheet prepared under US GAAP (the RWJ textbook format), how are the items ordered?

a. Assets in order of decreasing liquidity; liabilities in ascending order of when they are due
b. Assets in order of increasing liquidity; liabilities in descending order of when they are due
c. Non-current assets first, then current assets; non-current liabilities first
d. Assets and liabilities in order of size

10. Hanwoo Corp.'s balance sheet shows shareholders' equity of $80 million, while its 20 million shares trade at $12 each. Which statement is correct?

a. The market value of equity is $80 million
b. The market value of equity exceeds book value by $160 million
c. Book value is the better estimate of what shareholders would receive if they sold today
d. The difference implies an accounting error in the balance sheet

11. A bond's coupon rate is above the market yield, and the yield stays unchanged until maturity. What happens to its price over time?

a. It rises further above par as maturity approaches
b. It falls toward par and equals the face value at maturity
c. It stays at the same premium until the final coupon
d. It drops below par halfway through its life

12. You take a $250,000 mortgage for 30 years at 6%, compounded monthly. What is the monthly payment?

a. $1,513.52
b. $1,498.88
c. $8,994.73
d. $15,000.00

13. A stock sells for $50.00. Its next dividend will be $2.00, and dividends are expected to grow at 6% a year forever. What are its dividend yield, capital gains yield and required return?

a. 4.0%; 6.0%; 10.0%
b. 4.0%; 4.0%; 8.0%
c. 6.0%; 4.0%; 10.0%
d. 4.24%; 6.0%; 10.24%

14. Four bonds have the same yield to maturity. Which one's price is most sensitive to a change in interest rates?

a. A 2-year bond with a 10% coupon
b. A 30-year bond with a 10% coupon
c. A 2-year zero coupon bond
d. A 30-year zero coupon bond

15. Which feature shows that preferred stock is equity rather than debt?
    a. It pays a fixed periodic dividend
    b. It ranks ahead of common stock in liquidation
    c. It is often callable by the issuer
    d. Skipping a preferred dividend is not a default and cannot trigger bankruptcy

16. In computing cash flow from assets, how is an increase in net working capital during the year treated?
    a. It is subtracted, because the firm tied up additional cash in working capital
    b. It is added, because the firm holds more liquid assets
    c. It is ignored, because NWC is a balance-sheet item
    d. It is added to operating cash flow as a source of cash

17. A bond offers a nominal return of 8% for one year, and expected inflation is 3%. What is the real rate of return?
    a. 5.00%
    b. 4.85%
    c. 11.24%
    d. 5.15%

18. A $1,000 bond pays a 6% coupon semiannually ($30 per coupon). Four months have passed since the last coupon, and the bond is quoted at $1,012. What is the invoice (dirty) price?
    a. $1,032.00
    b. $1,022.00
    c. $1,012.00
    d. $1,042.00

19. You put $250 into an account at the end of every month for 20 years. The account pays 4.8%, compounded monthly. What will you have at the end?
    a. $100,418.76
    b. $97,126.75
    c. $100,820.43
    d. $60,000.00

20. Which statement about using financial leverage to raise ROE is correct?
    a. Higher leverage raises ROE without any effect on risk
    b. Higher leverage raises ROE only when assets earn more than the after-tax cost of debt, and it increases the risk borne by shareholders
    c. Leverage changes ROA but leaves ROE unchanged
    d. Leverage raises ROE only for firms with high profit margins

21. A loss-making firm has sales of $12.00 per share. Comparable firms trade at 1.8 times sales. What is the estimated share price?

a. $6.67
b. $21.60
c. $13.80
d. $0.00

22. An investor in the 25% tax bracket considers a municipal bond yielding 4.5%. What is its equivalent taxable yield?

a. 3.38%
b. 6.00%
c. 5.63%
d. 4.50%

23. A firm issues 5,000 shares of common stock with a par value of $1 at $15 per share. How much is recorded as additional paid-in capital (paid-in surplus)?

a. $75,000
b. $70,000
c. $5,000
d. $80,000

24. A preferred share pays $1.25 every quarter forever. Investors require 8%, compounded quarterly. What is the share worth?

a. $15.63
b. $250.00
c. $60.66
d. $62.50

25. A $36,000 machine can be leased for 36 payments of $1,150, each due at the beginning of the month. What annual rate, compounded monthly, does the lease charge?

a. 0.82%
b. 10.33%
c. 9.31%
d. 9.87%

26. Why is interest expense not deducted when computing operating cash flow?

a. Because interest is a noncash expense
b. Because interest is a financing cost paid to creditors, not an operating cost
c. Because interest is already included in depreciation
d. Because interest is not tax-deductible

27. Holding the quoted APR constant, what happens to the effective annual rate as compounding becomes more frequent?

a. It falls
b. It stays equal to the APR
c. It rises
d. It rises only if the APR exceeds 10%

28. A bond's coupon rate is above the current yield to maturity on comparable bonds. How does it sell?

a. At par
b. At a discount
c. At a premium
d. It cannot be determined without the maturity

29. Why does net income generally differ from operating cash flow?

a. Net income is measured before taxes, whereas OCF is measured after taxes
b. Net income subtracts noncash charges such as depreciation and also subtracts interest, a financing cost
c. Net income excludes cost of goods sold, whereas OCF includes it
d. Net income uses cash-basis accounting, whereas OCF uses accrual accounting

30. A firm reports sales of $850,000, costs of $610,000, depreciation of $95,000 and interest expense of $32,000. The tax rate is 21%. What is net income?

a. $113,000
b. $89,270
c. $114,550
d. $145,000

31. An investment pays $2,000 at the end of quarter 1, $3,000 at the end of quarter 2, nothing in quarter 3 and $4,000 at the end of quarter 4. The required return is 8%, compounded quarterly. What is it worth today?

a. $8,539.67
b. $8,613.58
c. $7,363.99
d. $9,000.00

32. A $1,000 bond with a 7% annual coupon and 10 years to maturity is priced at $1,075, which implies a YTM of 5.98%. What is its current yield, and how do the three yield measures rank?

a. Current yield 6.51%; YTM < current yield < coupon rate
b. Current yield 6.51%; coupon rate < current yield < YTM
c. Current yield 7.53%; YTM < coupon rate < current yield
d. Current yield 7.00%; all three are equal

33. On a common-size balance sheet and a common-size income statement, every item is expressed as a percentage of:

a. Total equity and net income
b. Total assets and net income
c. Total assets and sales
d. Sales and total assets

34. You save $300 at the end of every month in an account paying 7.2%, compounded monthly. How many years until the balance reaches $50,000?

a. 115.87 years
b. 9.97 years
c. 13.89 years
d. 9.66 years

35. Which of the following is NOT a reason why a stock is harder to value than a bond?

a. Dividends are paid at the board's discretion, while coupons are contractual
b. A stock has no maturity date
c. Both the required return and the growth rate must be estimated
d. A stock's value is not the present value of its future cash flows

36. Why does a call provision lower a bond's value to investors?

a. Because the call premium is paid to the issuer
b. Because the issuer calls the bond when rates have fallen, exactly when holders would like to keep the high coupon, so investors demand a higher yield
c. Because callable bonds cannot be traded in the secondary market
d. Because a call provision raises the bond's default risk

37. A firm reports net income of $640,000 with 200,000 shares outstanding and pays total dividends of $240,000. What are earnings per share, dividends per share and the addition to retained earnings?

a. $3.20; $1.20; $400,000
b. $3.20; $1.20; $640,000
c. $2.00; $1.20; $400,000
d. $3.20; $0.375; $400,000

38. Expected inflation rises, while nothing changes about an issuer's creditworthiness or the real return investors require. What happens to the issuer's outstanding fixed-coupon bonds?

a. Their prices rise because nominal coupons are worth more
b. Their prices are unaffected because default risk is unchanged
c. Nominal yields rise with expected inflation (Fisher effect), so their prices fall
d. Their coupon rates are reset upward

39. A company leases equipment worth $60,000. Payments are due at the beginning of each month for 3 years, and the lessor charges 9%, compounded monthly. What is the monthly lease payment?

a. $1,907.98
b. $5,654.10
c. $1,893.78
d. $1,666.67

40. Which rating is the lowest investment-grade rating?

a. Aa / AA
b. A / A
c. Baa3 / BBB−
d. Ba1 / BB+

41. Which statement is NOT one of the cautions about financial ratio analysis?

a. Firms may use different accounting methods such as FIFO or LIFO
b. Different fiscal year-ends mean balance sheets are snapshots on different dates
c. Each industry has a single theoretically correct value for every ratio
d. Firms may window-dress items around reporting dates

42. You want $1,000,000 in 25 years. Your account pays 6%, compounded monthly. How much must you deposit at the end of every month?

a. $1,518.89
b. $1,435.83
c. $1,443.01
d. $3,333.33

43. In the dividend growth model $P_0 = D_1 / (R - g)$, what happens if the assumed growth rate g is greater than or equal to R?

a. The share is worth an infinite amount, which is a realistic result for growth firms
b. The formula breaks down (negative or infinite price), signalling that such growth cannot last forever
c. The model still works if D_0 is used instead of D_1
d. The price equals D_1 / g

44. Which statement about tax rates is correct?

a. The average tax rate is the one used for investment decisions
b. Under the U.S. flat 21% corporate tax, the marginal and average rates are equal at every income level
c. Under a progressive schedule the average rate exceeds the marginal rate
d. The marginal rate is total taxes divided by taxable income

45. A firm has a total debt ratio of 0.60. What are its debt-equity ratio and equity multiplier?
a. D/E 1.50; EM 2.50
b. D/E 0.60; EM 1.60
c. D/E 2.50; EM 1.50
d. D/E 1.50; EM 1.67

46. You deposit $500 at the beginning of every month for 15 years. The account pays 6%, compounded monthly. What will you have at the end of 15 years?
a. $146,136.40
b. $145,409.36
c. $148,035.17
d. $90,000.00

47. In a nonconstant-growth valuation, dividends become constant-growth after date t, and the terminal price $P_t = D_{t+1} / (R - g)$ is computed. By how many periods should P_t be discounted to get today's value?
a. t + 1 periods, because it is based on D_{t+1}
b. t periods, because P_t is a price at date t
c. One period, because the model already discounts the dividends
d. It should not be discounted; it is added to P_0 directly

48. Which of the following is a positive covenant in a bond indenture?
a. Limit the dividends the firm may pay
b. Prohibit mergers without creditor consent
c. Maintain working capital above a stated level
d. Forbid the issue of additional long-term debt

49. Which statement is correct?
a. The firm receives cash every time its shares trade on the NYSE
b. In the secondary market investors trade existing shares and the firm receives nothing
c. In the primary market investors trade shares among themselves
d. An IPO is a secondary-market transaction

50. Why does a balance grow faster under compound interest than under simple interest?
a. Because the interest rate rises each year
b. Because interest is earned on previously earned interest as well as on the principal
c. Because simple interest is paid only at maturity
d. Because compounding uses the APR instead of the EAR

51. Which component of a corporate bond's nominal yield is bond-specific rather than part of the term structure?
a. Real rate
b. Inflation premium
c. Interest rate risk premium
d. Default risk premium

52. You deposit $1,000 at the beginning of every quarter into an account paying 8%, compounded quarterly. How many years until the balance reaches $100,000?

a. 54.81 years
b. 13.70 years
c. 13.87 years
d. 25.00 years

53. Coca-Cola and PepsiCo both earn an ROE of about 40%. According to the DuPont breakdown in the lecture, which statement is correct?

a. Coca-Cola gets there through a high profit margin; PepsiCo relies on faster asset turnover and greater financial leverage
b. PepsiCo gets there through a high profit margin; Coca-Cola relies on leverage
c. Both firms have nearly identical margins, turnover and leverage
d. Coca-Cola has the higher asset turnover because it owns more plants

54. You want to earn an effective annual rate of 8% with monthly compounding. What APR must you be quoted?

a. 8.00%
b. 7.72%
c. 8.30%
d. 0.64%

55. A dealer quotes a stock at bid $49.90 / ask $50.10. Which statement is correct?

a. You can sell to the dealer at $49.90 and buy from the dealer at $50.10; the $0.20 spread is the dealer's compensation
b. You can buy at $49.90 and sell at $50.10
c. A broker would quote the same prices and hold the stock in inventory
d. The spread is a commission the dealer pays to the exchange

56. Why does a bond buyer pay the seller accrued interest on top of the quoted (clean) price?

a. Because the dealer adds a commission to every trade
b. Because the next coupon is paid in full to whoever holds the bond on the payment date, so the buyer reimburses the seller for the part the seller earned
c. Because accrued interest is a tax collected at settlement
d. Because the clean price already includes the next coupon

57. A firm reports negative cash flow from assets for the year. Which interpretation is most appropriate?

a. The firm is necessarily in financial distress
b. The firm must have reported a net loss
c. The firm raised funds externally during the year, possibly to finance growth
d. The firm's cash flow to creditors must be negative

58. In a loan repaid with equal payments, how does each payment change over time?
a. The interest portion rises and the principal portion falls
b. The payment is constant, the interest portion falls and the principal portion rises
c. Both the interest and principal portions are constant
d. The payment falls each period as the balance falls

59. Which statement about NASDAQ is correct?
a. It is an auction market organized around designated market makers on a trading floor
b. It is a dealer market: a computer network in which several market makers compete in the same stock
c. It has no market makers; all trades are matched between customers
d. It lists only preferred stock

60. Why is inventory subtracted from current assets in the quick (acid-test) ratio?
a. Because inventory is not a current asset
b. Because inventory is the least liquid current asset and its book value is the least reliable
c. Because inventory is already counted in cash
d. Because inventory is financed with long-term debt

61. Which of the following is NOT one of the assumptions behind a bond's yield to maturity?
a. The bond is held to maturity
b. There is no default
c. Coupons are reinvested at the YTM
d. Coupons are reinvested at the coupon rate

62. You deposited $200 at the end of every month for 10 years, and the account is now worth $40,000. What annual rate, compounded monthly, did it earn?
a. 9.58%
b. 0.80%
c. 10.01%
d. 6.67%

63. Which statement about the tax treatment of financing costs is correct?
a. Both interest and dividends are tax-deductible to the paying corporation
b. Interest is tax-deductible to the corporation; dividends are not
c. Dividends are tax-deductible to the corporation; interest is not
d. Neither affects the corporation's tax bill

64. Your budget allows a car payment of $520 per month for 5 years. The bank charges 9.6%, compounded monthly. How large a loan can you take?
a. $5,394.53
b. $2,538.75
c. $31,200.00
d. $24,702.19

65. A firm skipped two years of dividends on its cumulative preferred stock. Which statement is correct?

a. The firm is in default and the preferred holders can force bankruptcy
b. The arrears must be paid before any dividend is paid on common stock
c. The skipped dividends are forgiven when the board resumes payments
d. Preferred holders immediately become creditors

66. A loan is quoted at 12%, compounded quarterly. What is the effective annual rate?

a. 12.55%
b. 12.00%
c. 3.00%
d. 12.68%

67. PepsiCo's current ratio is 0.85. According to the lecture, how should this be read?

a. As a clear sign of imminent financial distress
b. Not necessarily a red flag: steady cash sales fund working capital, long supplier terms inflate payables, and strong credit lets it roll short-term debt
c. As evidence that PepsiCo holds too much idle cash
d. As proof that PepsiCo's ROE must be lower than Coca-Cola's

68. What is the value of a firm that is not paying dividends today?

a. Zero, because the dividend discount model only values dividends already paid
b. It has value only because it is expected to pay dividends (or be bought by someone who will) at some point in the future
c. It equals its book value per share
d. It can only be valued with the P/E ratio of the same firm

69. In the short run, which of the following is best described as a fixed cost?

a. Raw materials consumed in production
b. Sales commissions paid per unit sold
c. A lease payment on a warehouse under a five-year contract
d. Hourly wages of temporary production workers

70. How does a put provision affect a bond compared with an otherwise identical bond without one?

a. It lowers the bond's value because the issuer can buy it back early
b. It raises the bond's value because the holder may force the issuer to buy it back at a stated price
c. It has no effect on value
d. It converts the bond into equity

Business Finance Midterm — Answers

1. b [Exhibit A]
EPS = 948 / 100 = 9.48, so P/E = 74 / 9.48 = 7.81
sales per share = 4,800 / 100 = 48, so P/S = 74 / 48 = 1.54
book value per share = 3,700 / 100 = 37, so M/B = 74 / 37 = 2.00.

2. b [Exhibit A]
TIE = EBIT / interest = 1,400 / 200 = 7.00
cash coverage = (EBIT + depreciation) / interest = 1,900 / 200 = 9.50.

3. b [Exhibit A]
Common-size balance sheet items are divided by total assets: 900 / 6,400 = 14.06%
common-size income statement items are divided by sales: 948 / 4,800 = 19.75%.

4. a [Exhibit A]
EV = 7,400 + 2,700 − 200 = 9,900 (total liabilities = 6,400 − 3,700 = 2,700)
EBITDA = 1,400 + 500 = 1,900
EV/EBITDA = 9,900 / 1,900 = 5.21.

5. a [Exhibit A]
EPS = 948 / 100 = 9.48
price = EPS × benchmark P/E = 9.48 × 12 = $113.76
BA II Plus: 948 ÷ 100 = 9.48, × 12 = 113.76.

6. a [Exhibit A]
Inventory turnover = COGS / inventory = 2,900 / 900 = 3.22
receivables turnover = sales / AR = 4,800 / 700 = 6.86
total asset turnover = sales / total assets = 4,800 / 6,400 = 0.75.

7. d [Exhibit A]
Accounts receivable rose by 100 (an increase in an asset is a use of cash)
accounts payable rose by 40 (an increase in a liability is a source of cash)
net effect = −100 + 40 = $60 use of cash.

8. b
The coupon rate (annual coupon ÷ face value) is fixed when the bond is issued
the YTM is the market's required return and changes daily, so the price is whatever makes the two consistent.

9. a
GAAP lists the most liquid assets first and the liabilities due soonest first
IFRS / K-IFRS flips the order and lists non-current items first, with the same financial logic.

10. b
Market value of equity = 20 million × $12 = $240 million versus book value of $80 million, a difference of $160 million
book values rest on historical cost, so a gap is normal and market value is what matters for decisions.

11. b
The premium is given back to the holder as a capital loss over the remaining life, so the price drifts down to par and equals the face value at maturity.

12. b
r = 0.5% per month, t = 360: 250,000 = C × [1 − 1/(1.005)^{360}] / 0.005, so C = $1,498.88 (360 N, 0.5 I/Y, 250000 PV, 0 FV, CPT PMT).

13. a
Dividend yield = D_1 / P_0 = 2.00 / 50.00 = 4.0%
under constant growth the price grows at g, so the capital gains yield = 6.0%
R = 4.0% + 6.0% = 10.0%
BA II Plus: 2 ÷ 50 = 0.04, + 0.06 = 0.10.

14. d
Interest rate risk is greater for a longer maturity and for a lower coupon
a zero coupon bond is the extreme case because all of its value sits at maturity, so the 30-year zero is the most rate-sensitive.

15. d
Preferred stock looks like debt (fixed dividend, no vote, ranks ahead of common, often callable), but missing a dividend is not a default, the dividend is not tax-deductible and holders rank behind every creditor, so it is equity.

16. a
An increase in NWC is a cash outflow: the firm invested cash in current assets net of current liabilities, so CFFA = OCF − NCS − ΔNWC.

17. b
Fisher: 1 + R = (1 + r)(1 + h), so r = 1.08 / 1.03 − 1 = 4.85%
BA II Plus: 1.08 ÷ 1.03 = 1.04854, − 1 = 0.0485.

18. a
Accrued interest = 30 × (4 / 6) = 20
dirty price = clean price + accrued interest = 1,012 + 20 = $1,032.00
BA II Plus: 30 × 4 ÷ 6 = 20, + 1012 = 1,032.

19. a
r = 0.4%, t = 240: FV = 250 × [(1.004)^{240} − 1] / 0.004 = $100,418.76 (240 N, 0.4 I/Y, −250 PMT, 0 PV, CPT FV).

20. b
Leverage enters ROE through the equity multiplier (ROE = ROA × EM)
it magnifies returns in both directions, so it raises ROE when assets earn more than debt costs and lowers it further when they earn less.

21. b
Price = sales per share × benchmark price-sales ratio = 12.00 × 1.8 = $21.60
BA II Plus: 12 × 1.8 = 21.60.

22. b
Equivalent taxable yield = muni yield / (1 − t) = 4.5% / 0.75 = 6.00%
BA II Plus: 4.5 ÷ 0.75 = 6.00.

23. b
Paid-in surplus = (issue price − par) × shares = (15 − 1) × 5,000 = $70,000
the $5,000 of par goes to the common stock line.

24. d
The dividend and the rate must be on the same basis: r = 8% / 4 = 2% per quarter
PV = C / r = 1.25 / 0.02 = $62.50
BA II Plus: 1.25 ÷ 0.02 = 62.50.

25. d
BGN mode: 36 N, 36000 PV, −1150 PMT, 0 FV, CPT I/Y = 0.8227% per month
quoted rate = 0.8227 × 12 = 9.87%
switch BGN off afterwards
Formula: $36{,}000 = 1{,}150 \times [1 - 1/(1 + r)^{36}] / r \times (1 + r)$, solved by iteration.

26. b
OCF = EBIT + depreciation − taxes measures the cash generated by operations before financing
interest belongs to the cash flow to creditors.

27. c
More frequent compounding pays interest on interest more often, so the EAR = $(1 + \text{APR}/m)^m - 1$ rises with m (15% APR: 15.00% annual, 16.08% monthly, 16.18% daily).

28. c
If the coupon rate exceeds the yield, the coupon is generous in the market and investors bid the price above face value
the premium is given back as a capital loss over the remaining life.

29. b
OCF = EBIT + depreciation − taxes adds depreciation back and leaves interest out, because it measures the cash generated by operations before financing net income is an accrual figure after both.

30. b
EBIT = 850,000 − 610,000 − 95,000 = 145,000
taxable income = 145,000 − 32,000 = 113,000
taxes = 0.21 × 113,000 = 23,730
net income = 113,000 − 23,730 = $89,270.

31. a
r = 2% per quarter: PV = $2{,}000/1.02 + 3{,}000/(1.02)^2 + 0/(1.02)^3 + 4{,}000/(1.02)^4$ = $8,539.67
BA II Plus: CF, CF0 = 0, C01 = 2000, C02 = 3000, C03 = 0, C04 = 4000, NPV I = 2, CPT = 8,539.67.

32. a
Current yield = 70 / 1,075 = 6.51%
for a premium bond the ordering is the reverse of a discount bond: YTM (5.98%) < current yield (6.51%) < coupon rate (7.00%)
BA II Plus: 70 ÷ 1075 = 0.0651
the YTM comes from 10 N, −1075 PV, 70 PMT, 1000 FV, CPT I/Y = 5.98.

33. c
Common-size balance sheets divide every account by total assets
common-size income statements divide every line by sales, which allows comparison over time and across firms of different sizes.

34. d
r = 0.6%: 0.6 I/Y, −300 PMT, 0 PV, 50000 FV, CPT N = 115.87 months
115.87 / 12 = 9.66 years
Formula: $50{,}000 = 300 \times [(1.006)^t - 1] / 0.006$, t in months.

35. d
The discounting principle is identical for stocks and bonds: the value of any financial asset is the PV of the cash flows it delivers
what is harder is the inputs (discretionary dividends, no maturity, estimated R and g).

36. b
The call option belongs to the issuer, who uses it when rates fall
investors are left reinvesting at lower rates, so they require a higher yield (a lower price) on a callable bond. A put provision is the mirror image and raises a bond's value.

37. a
EPS = 640,000 / 200,000 = $3.20
DPS = 240,000 / 200,000 = $1.20
addition to retained earnings = 640,000 − 240,000 = $400,000.

38. c
Bond yields are nominal: 1 + R = (1 + r)(1 + h), so a higher h raises the required nominal yield and lowers the price of a fixed-coupon bond.

39. c
r = 0.75%, t = 36: 60,000 = C × [1 − 1/(1.0075)^{36}] / 0.0075 × 1.0075, so C = $1,893.78 (BGN mode: 36 N, 0.75 I/Y, 60000 PV, 0 FV, CPT PMT
switch BGN off afterwards).

40. c
Investment grade runs from Aaa/AAA down to Baa3/BBB−
anything below is speculative grade (junk). Ratings measure default risk, not interest rate risk.

41. c
There is no underlying theory giving a single correct value
a ratio needs a benchmark (time-trend or peer-group analysis), and comparisons can be distorted by accounting methods, fiscal year-ends, conglomerates, capital structure, seasonality and window dressing.

42. c
r = 0.5%, t = 300: 1,000,000 = C × [(1.005)^{300} − 1] / 0.005, so C = $1,443.01 (300 N, 0.5 I/Y, 0 PV, 1000000 FV, CPT PMT).

43. b
The model requires R > g
if g ≥ R the algebra returns a negative or infinite value, which means the growth assumption is impossible to sustain indefinitely. The g in the model is the rate the firm can sustain forever.

44. b
The marginal rate (on the next dollar) is used for decisions
with a flat rate marginal = average, and with a progressive schedule (e.g. Korea) the marginal rate is above the average rate.

45. a
Equity is 0.40 of assets: EM = TA / TE = 1 / 0.40 = 2.50
D/E = 0.60 / 0.40 = 1.50, and EM = 1 + D/E.

46. a
r = 0.5%, t = 180: ordinary FV = 500 × [(1.005)^{180} − 1] / 0.005 = 145,409.36
annuity due = 145,409.36 × 1.005 = $146,136.40
BA II Plus: 180 N, 0.5 I/Y, −500 PMT, 0 PV, CPT FV = 145,409.36, × 1.005 = 146,136.40 (or BGN mode switch BGN off afterwards).

47. b
P_t is the value at date t of all dividends from t + 1 onward, so it is discounted by $(1 + R)^t$
discounting by t + 1 is the most common mistake.

48. c
Positive covenants state what the firm must do (maintain working capital or ratios, furnish audited statements, keep collateral in good condition)
the others are negative covenants. Firms accept covenants because the lower yield is worth more than the lost flexibility.

49. b
In the primary market the issuer sells securities for the first time and raises money
in the secondary market existing shares change hands between investors, and the firm receives nothing.

50. b
Simple interest is paid only on the original principal
compound interest pays interest on principal and accumulated interest, so FV = PV × $(1 + r)^t$ grows faster and faster.

51. d
The term structure (real rate + inflation premium + interest rate risk premium) is the same for every bond of a given maturity
the default risk, taxability and liquidity premiums are specific to each bond.

52. b
r = 2% per quarter, BGN mode: 2 I/Y, −1000 PMT, 0 PV, 100000 FV, CPT N = 54.81 quarters
54.81 / 4 = 13.70 years
switch BGN off afterwards
Formula: 100,000 = 1,000 × [(1.02)^t − 1] / 0.02 × 1.02, t in quarters.

53. a
KO: profit margin 27.34%, asset turnover 0.46, equity multiplier 3.26
PEP: 8.77%, 0.87, 5.26. Same ROE, different routes: KO's asset-light concentrate model gives high margins.

54. b
APR = 12 × [(1.08)$^{1/12}$ − 1] = 12 × 0.6434% = 7.72%
(ICONV: EFF = 8, C/Y = 12, CPT NOM).

55. a
The bid is the price the dealer pays (you sell), the ask is the price the dealer accepts (you buy), and the spread compensates the dealer for making a market and bearing inventory risk
a broker only matches trades for a commission.

56. b
The dirty (invoice) price = clean price + accrued interest, where accrued interest = coupon × days since last coupon ÷ days in the coupon period.

57. c
Negative CFFA means operations plus asset purchases used more cash than they generated, so the gap was financed by new debt or equity
this is routine for fast-growing firms making heavy investments.

58. b
Interest each period = rate × beginning balance
as the balance falls, interest falls and, with a constant payment, the principal portion rises.

59. b
NASDAQ has no physical floor
trades are executed against competing market makers, whose competition narrows the spread. The NYSE is a hybrid auction market with DMMs, now largely electronic.

60. b
Quick ratio = (current assets − inventory) / current liabilities
inventory may not be sold quickly at book value, especially when a firm is in trouble.

61. d
YTM is the bond's internal rate of return
it is a promised yield that assumes the bond is held to maturity, no default occurs and coupons are reinvested at the YTM itself.

62. a
120 N, −200 PMT, 0 PV, 40000 FV, CPT I/Y = 0.7984% per month
quoted rate = 0.7984 × 12 = 9.58% (the EAR would be 10.01%)
Formula: $40{,}000 = 200 \times [(1 + r)^{120} - 1] / r$, solved by iteration.

63. b
Interest is deducted before computing taxable income
dividends are paid out of after-tax profit, which gives debt a tax advantage at the corporate level.

64. d
r = 9.6% / 12 = 0.8% per month, t = 60: PV = $520 \times [1 - 1/(1.008)^{60}] / 0.008$ = $24,702.19 (60 N, 0.8 I/Y, −520 PMT, 0 FV, CPT PV).

65. b
Cumulative preferred dividends accumulate as arrearages
no common dividend can be paid until they are cleared, but skipping them is not a default.

66. a
EAR = $(1 + 0.12/4)^4 - 1 = (1.03)^4 - 1$ = 12.55% (ICONV: NOM = 12, C/Y = 4, CPT EFF).

67. b
A current ratio below 1 can be normal for a firm with steady cash sales, large payables and cheap access to short-term credit
a big current-asset cushion would only sit idle.

68. b
The price of a share is the PV of all future dividends
a firm paying nothing now must be expected to pay eventually, otherwise its shares are worthless.

69. c
A fixed cost does not vary with output over the short-run horizon
a contractual lease payment must be paid whatever the output level.

70. b
A put gives the option to the holder, the mirror image of a call provision, so it raises the bond's value (lowers its yield).

Chapter 3: Working with Financial Statements

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. Harborview Outfitters has market capitalization of $20,000 thousand (500 thousand shares at $40), interest-bearing debt of $1,760 thousand (notes payable plus long-term debt), cash of $260 thousand and EBITDA of $1,530 thousand. What is EV/EBITDA?

a. 13.07

b. 14.22

c. 14.05

d. 14.39

2. Firm X has a profit margin of 2%, total asset turnover of 4.0 and an equity multiplier of 2.5. Firm Y has a profit margin of 8%, total asset turnover of 1.0 and an equity multiplier of 2.5. Which statement is correct?

a. Firm X relies more on financial leverage than Firm Y.

b. Firm Y uses its assets more efficiently than Firm X.

c. Both earn an ROE of 20%: X through rapid asset turnover and Y through high margins, with equal leverage.

d. Firm Y earns the higher ROE because its margin is higher.

Chapter 3: Working with Financial Statements — Answers

1. c
Step 1: EV = Market cap + Interest-bearing debt − Cash
Step 2: EV = 20,000 + 1,760 − 260 = 21,500
Step 3: EV/EBITDA = 21,500 / 1,530 = 14.05
Why subtract cash: an acquirer also acquires the firm's cash.

2. c
Formula: ROE = Profit margin × Total asset turnover × Equity multiplier
Step 1: ROE(X) = 0.02 × 4.0 × 2.5 = 20%
Step 2: ROE(Y) = 0.08 × 1.0 × 2.5 = 20%
Step 3: Equity multipliers are equal (2.5), so leverage is the same
Reading: X turns assets over 4 times faster, Y earns a 4 times higher margin.

Chapter 5: Discounted Cash Flow Valuation

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. A lease requires $1,000 at the beginning of every month for 24 months. The rate is 6% compounded monthly. What is the lease worth today?

a. $22,562.87
b. $24,000.00
c. $22,675.68
d. $22,450.61

2. Lender A quotes 8.4% compounded monthly. Lender B quotes 8.5% compounded semiannually. Which loan is cheaper?

a. Lender A, because its quoted rate is lower
b. Lender A, because its EAR is 8.68% versus 8.73% for B
c. Lender B, because its EAR is 8.68% versus 8.73% for A
d. They cost the same

3. You save $400 at the end of every month in an account paying 6% compounded monthly. How many years until the balance reaches $30,000?

a. 63.85 years
b. 6.25 years
c. 5.32 years

4. You will receive $5,000 a year for 10 years, but the first payment arrives 4 years from today. The discount rate is 8%. What is the stream worth today?

a. $33,550.41
b. $24,660.55
c. $26,633.39
d. $28,764.07

5. You borrow $20,000 for 4 years with monthly payments. The bank quotes 8% compounded quarterly. What is the monthly payment?

a. $488.26
b. $416.67
c. $652.04
d. $487.76

Chapter 5: Discounted Cash Flow Valuation — Answers

1. c
Step 1: Monthly rate r = 6% / 12 = 0.5%
Step 2: PV (ordinary) = 1,000 × [1 − 1/(1.005)^{24}] / 0.005 = 22,562.87
Step 3: PV (due) = 22,562.87 × 1.005 = 22,675.68
BA II Plus: 24 N, 0.5 I/Y, −1000 PMT, 0 FV, CPT PV = 22,562.87
Then: × 1.005 = 22,675.68 (or BGN mode).

2. c
Step 1: EAR(A) = (1 + 0.084/12)^{12} − 1 = 8.73%
Step 2: EAR(B) = (1 + 0.085/2)^{2} − 1 = 8.68%
Step 3: Compare EARs, never quoted rates
Answer: B is cheaper (lower EAR).

3. c
Step 1: 30,000 = 400 × [(1.005)^{t} − 1] / 0.005
Step 2: (1.005)^{t} = 1.375
Step 3: t = ln 1.375 / ln 1.005 = 63.85 months
Step 4: Years = 63.85 / 12 = 5.32
BA II Plus: 0.5 I/Y, −400 PMT, 0 PV, 30000 FV, CPT N = 63.85 (months).

4. c
Step 1: The annuity formula gives the value one period before the first payment, at date 3
Step 2: PV_3 = 5,000 × [1 − 1/(1.08)^{10}] / 0.08 = 33,550.41
Step 3: PV_0 = 33,550.41 / (1.08)^{3} = 26,633.39
Watch for: discounting 4 years is the common error
BA II Plus: 10 N, 8 I/Y, −5000 PMT, 0 FV, CPT PV = 33,550.41
then 3 N, 0 PMT, −33550.41 FV, CPT PV = 26,633.39.

5. d
Step 1: Quarterly rate = 8% / 4 = 2%
Step 2: Effective monthly rate = (1.02)$^{1/3}$ − 1 = 0.6623%
Step 3: 20,000 = C × [1 − 1/(1.006623)^{48}] / 0.006623
Step 4: C = $487.76
Watch for: 8% / 12 assumes monthly compounding and gives $488.26
BA II Plus: 48 N, 0.6623 I/Y, 20000 PV, 0 FV, CPT PMT = −487.76.

Chapter 6: Interest Rates and Bond Valuation

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. A firm issued a bond at par with an 8% coupon. Market interest rates have since risen. Which statement is correct?

a. The coupon rate rises to match the market, so the price stays at par.
b. The coupon rate stays at 8%, the YTM rises, and the price falls below par.
c. The YTM stays at 8% because it is set at issue.
d. The face value falls to restore the 8% yield.

2. A $1,000 bond pays an 8% coupon semiannually and matures in 6 years. The yield to maturity is 10% (quoted annually). What is its price?

a. $591.18
b. $911.37
c. $912.89
d. $1,000.00

3. A $1,000 bond pays an 8% annual coupon, matures in 3 years and now sells at par. If its yield rises to 9%, by how much does its price change?

a. −1.00%
b. −2.53%
c. −3.00%
d. +2.62%

4. A bond's yield can either rise by 2 percentage points or fall by 2 percentage points. Which statement about the price change is correct?

a. The price changes by the same dollar amount in either direction.
b. The price falls more when yields rise than it rises when yields fall.
c. The price rises more when yields fall than it falls when yields rise.
d. The price does not change because the coupon is fixed.

5. Four bonds currently yield 8%. Which bond's price changes most when the market yield rises by one percentage point?

a. 5-year maturity, 10% coupon
b. 5-year maturity, 4% coupon
c. 20-year maturity, 10% coupon
d. 20-year maturity, 4% coupon

6. A bond's quoted yield to maturity equals the return an investor actually earns only under which conditions?

a. The bond is held to maturity, it does not default, and coupons are reinvested at the YTM.
b. The bond is sold before maturity at par.
c. Interest rates fall after purchase.
d. The coupon rate equals the current yield.

7. A $1,000 bond pays a 6% coupon semiannually, matures in 10 years and sells for $925. What is its yield to maturity (quoted annually)?

a. 3.53%
b. 6.49%
c. 7.06%
d. 7.18%

8. A bond with an 8% coupon rate and 8 years to maturity trades below par. Which ordering of its yields is correct?

a. YTM < current yield < coupon rate
b. Current yield < coupon rate < YTM
c. Coupon rate = current yield = YTM
d. Coupon rate < current yield < YTM

9. Compared with an otherwise identical non-callable bond, a callable bond should have which characteristics?

a. A higher price and a lower yield
b. A lower price and a higher yield
c. The same price and yield
d. A higher price and a higher yield

10. A municipal bond yields 4.2% and a comparable taxable bond yields 6.0%. Three investors face marginal tax rates of 20%, 30% and 35%. Which statement is correct?

a. All three prefer the taxable bond.

b. All three prefer the muni.

c. Only the 35% investor prefers the muni; the 30% investor is indifferent.

d. Only the 20% investor prefers the muni.

11. A $1,000 bond pays an 8% coupon semiannually. Two months have passed since the last coupon, and the bond is quoted at $985. What is the invoice (dirty) price?

a. $971.67

b. $985.00

c. $998.33

d. $1,025.00

12. A bond yields 11.3% and expected inflation is 6%. What is the exact real rate?

a. 5.30%

b. 5.00%

c. 17.98%

13. A $1,000 bond pays a 6% coupon semiannually and matures in 10 years. Investors require an effective annual return of 8.16%. What is the bond's price?

a. $1,000.00

b. $854.26

c. $864.10

d. $856.10

14. How does interest rate risk change with maturity?

a. It falls as maturity rises.

b. It rises with maturity at an increasing rate.

c. It rises with maturity, but at a decreasing rate.

d. It does not depend on maturity.

15. Among bonds with the same maturity, which is the most sensitive to interest rate changes?

a. A bond with a 15% coupon

b. A bond with a 10% coupon

c. A bond with a 5% coupon

d. A zero coupon bond

16. Expected inflation rises, but nothing changes about the issuer's credit or the real return investors require. What happens to the bond's price?

a. Nothing, because the coupon is fixed.

b. It rises, because the coupon is worth more.

c. It falls, because nominal yields rise with expected inflation.

d. It rises, because the real rate falls.

17. Compared with a Treasury bond of the same maturity, which premiums explain the higher yield on a corporate bond?

a. Real rate and inflation premium

b. Interest rate risk premium only

c. Default risk, taxability and liquidity premiums

18. Which pairing of bond type and feature is correct?

a. Put bond: lets the issuer buy the bond back, lowering its value

b. Convertible bond: the holder may exchange it for shares, so it carries a lower coupon

c. Floating-rate bond: the coupon is fixed for life

d. Income bond: coupons are paid even if the firm has no income

Chapter 6: Interest Rates and Bond Valuation — Answers

1. b
Coupon rate: set at issue and fixed for the life of the bond
YTM: set by the market and changes every day
Price: whatever makes the two consistent
Here: higher YTM with a fixed coupon means a lower price.

2. b
Step 1: Convert to periods: $t = 6 \times 2 = 12$
$C = 80 / 2 = 40$
$r = 10\% / 2 = 5\%$
Step 2: PV of coupons $= 40 \times [1 - 1/(1.05)^{12}] / 0.05 = 354.53$
Step 3: PV of face value $= 1{,}000 / (1.05)^{12} = 556.84$
Step 4: Price $= 354.53 + 556.84 = 911.37$
BA II Plus: 12 N, 5 I/Y, 40 PMT, 1000 FV, CPT PV = −911.37.

3. b
Step 1: At 8% the price is \$1,000 (coupon rate = YTM)
Step 2: At 9%: Price $= 80 \times [1 - 1/(1.09)^{3}] / 0.09 + 1{,}000 / (1.09)^{3} = 974.69$
Step 3: Change $= 974.69 / 1{,}000 - 1 = -2.53\%$
Rule: prices and yields move in opposite directions.

4. c
Shape: the price-yield curve is convex
Result: prices rise faster when yields fall than they drop when yields rise by the same amount
Note: interest rate risk is price risk, not default risk.

5. d
Maturity effect: a longer bond has more cash flows exposed, and the distant face value swings most
Coupon effect: a lower coupon pushes more value to maturity, so it is more sensitive
Answer: the 20-year, 4% coupon bond combines both effects.

6. a
YTM: the single discount rate that sets the PV of the promised cash flows equal to the price (the bond's IRR)
Assumption 1: held to maturity
Assumption 2: no default (YTM is a promised yield)
Assumption 3: coupons reinvested at the YTM
If rates fall, reinvested coupons earn less and the realized return falls short.

7. c
Step 1: $t = 20$ periods
$C = 30$
Price $= 925$
Step 2: Solve $925 = 30 \times [1 - 1/(1 + r)^{20}] / r + 1{,}000 / (1 + r)^{20}$
Step 3: $r = 3.53\%$ per six months
Step 4: Quoted YTM $= 3.53\% \times 2 = 7.06\%$
BA II Plus: 20 N, −925 PV, 30 PMT, 1000 FV, CPT I/Y = 3.53
Check: price below par, so YTM > coupon rate.

8. d
Coupon rate = Annual coupon / Face value
Current yield = Annual coupon / Price
it is higher because the price is below face
YTM: also counts the capital gain, so it is highest
Discount bond: coupon rate < current yield < YTM.

9. b
Step 1: The issuer calls when rates have fallen
Step 2: Holders lose the high coupon exactly when they would most like to keep it
Step 3: So investors demand a higher yield
Result: a higher yield means a lower price.

10. c
Formula: Equivalent taxable yield $= 4.2\% / (1 - t)$
20%: $4.2 / 0.80 = 5.25\% < 6.0\%$, taxable is better
30%: $4.2 / 0.70 = 6.00\%$, indifferent
35%: $4.2 / 0.65 = 6.46\% > 6.0\%$, muni is better
Lesson: munis are held mainly by high-bracket investors.

11. c
Step 1: Coupon per period $= 80 / 2 = 40$
Step 2: Accrued interest $= 40 \times (2 / 6) = 13.33$
Step 3: Dirty price = Clean price + Accrued interest $= 985.00 + 13.33 = 998.33$
Why: the buyer reimburses the seller for the part of the next coupon the seller earned.

12. b
Formula: $1 + R = (1 + r) \times (1 + h)$
Step 1: $1 + r = 1.113 / 1.06 = 1.05$
Step 2: $r = 5.00\%$
Note: the approximation $R - h = 5.30\%$ is not exact, and the error grows with inflation.

13. c
Step 1: Semiannual rate $= (1.0816)^{1/2} - 1 = 4.00\%$
Step 2: $t = 20$
$C = 30$
Step 3: Price $= 30 \times [1 - 1/(1.04)^{20}] / 0.04 + 1{,}000 / (1.04)^{20} = 864.10$
Watch for: $8.16\% / 2 = 4.08\%$ treats an effective rate as a quoted rate and gives \$854.26
BA II Plus: 20 N, 4 I/Y, 30 PMT, 1000 FV, CPT PV = −864.10.

14. c
Rule: longer bonds are hit harder by rate changes
But: the effect rises at a decreasing rate
Example: going from 1 to 10 years matters far more than going from 20 to 30 years.

15. d
Rule: for a given maturity, the lower the coupon, the greater the price sensitivity
Reason: a low-coupon bond delivers most of its value at maturity, far in the future
Extreme case: a zero coupon bond has all of its value at maturity.

16. c
Bond yields are nominal
Fisher effect: higher expected inflation raises nominal yields
Higher yield with fixed promised cash flows means a lower price.

17. c
Term structure: real rate, inflation premium and interest rate risk premium are common to both bonds
Bond-specific: default risk, taxability and liquidity premiums
A corporate bond carries these and the Treasury largely does not.

18. b
Convertible: exchangeable for a fixed number of shares at the holder's option
the right is valuable, so the coupon is lower
Put bond: the holder may force the issuer to buy it back
it raises the bond's value
Floater: coupon resets against a reference rate, often with a floor and ceiling (collar)
Income bond: coupons paid only if income is sufficient.

Chapter 7: Equity Markets and Stock Valuation

Multiple Choice · Chapters 3, 5–7 · 55 questions · ★ = harder

1. A company just paid a dividend of $2.00. Dividends grow at 5% a year forever and the required return is 12%. What will the stock be worth five years from now?
   a. $36.47
   b. $30.00
   c. $38.29
   d. $40.20

2. The required return on a stock is 9%. The next dividend is $1.00 and the growth rate is assumed to be 10% forever. What does the dividend growth model say?
   a. The price is $100, so the stock is undervalued.
   b. The price is $11.11 because growth is ignored when g exceeds R.
   c. The price is −$100, which shows the assumption g > R cannot hold forever.

3. A firm just paid a dividend of $2.00. Dividends grow at 15% a year for the next 3 years and at 5% a year forever after that. The required return is 12%. What is the stock worth today?
   a. $30.00
   b. $35.32
   c. $38.80
   d. $45.63

4. A firm will pay no dividend for the next 3 years. It will pay $1.20 at the end of year 4, and that dividend will then grow at 5% forever. The required return is 13%. What is the stock worth today?

a. $15.00
b. $9.20
c. $10.40
d. $0.74

5. Which pairing of growth case and its watch-out is correct?

a. Zero growth: $D_1 = D_0$ holds in every case.
b. Constant growth: use D_0 in the numerator.
c. Nonconstant growth: discount P_t by t + 1 periods.
d. Constant growth: use D_1 and require R > g.

6. Which statement about preferred stock is correct?

a. Failing to pay a preferred dividend triggers default and bankruptcy.
b. Preferred dividends are tax-deductible to the issuer.
c. Preferred holders rank ahead of all creditors in liquidation.
d. Preferred dividends are not tax-deductible, and skipping one is not a default.

7. A firm just paid a dividend of $2.00. Dividends grow at 8% a year for 5 years and at 4% a year forever after that. The required return is 10%. What is the stock worth today?

a. $34.67
b. $9.47
c. $50.94
d. $41.10

8. A stock sells for $50. It just paid a dividend of $2.00, and investors require 11%. What constant growth rate is the market implying?

a. 4.00%
b. 7.00%
c. 6.73%
d. 5.50%

9. A stock sells for $36. It just paid a dividend of $1.80, and dividends grow at 5% forever. What is the required return?

a. 10.00%
b. 10.25%
c. 5.25%
d. 15.00%

10. A firm earns an ROE of 15% and pays out 40% of its earnings as dividends. Next year's earnings per share are expected to be $3.00, and investors require 13%. What is the stock worth?

a. $12.86
b. $20.00
c. $30.00
d. $13.33

11. The next dividend is $2.00 and dividends grow at 5% forever. If the required return rises from 10% to 11%, by how much does the stock price change?

a. −1.0%
b. −9.1%
c. −16.7%
d. −50.0%

12. A stock sells for $60. Investors require 12% and dividends grow at 8% forever. What dividend did the firm just pay?

a. $2.40
b. $4.80
c. $2.22
d. $7.20

13. A firm pays out 50% of its earnings. Dividends grow at 6% forever and the required return is 12%. What price-earnings ratio (price ÷ next year's earnings) does the dividend growth model imply?

a. 6.00
b. 16.67
c. 4.17
d. 8.33

14. Han River Utilities has preferred stock with a $100 par value and an annual dividend rate of 7.5%, paid quarterly. The dividend never changes. Investors require 9%, compounded monthly. What is the price of one preferred share?

a. $83.33
b. $82.71
c. $79.95
d. $250.00

15. Sejong Holdings has just issued preferred stock that pays a fixed $4.50 per year forever, but the first dividend will not be paid until three years from today. The required return is 12%. What is the price today?

a. $26.69
b. $33.48
c. $37.50
d. $29.89

16. Daehan Foods just paid a dividend of $1.80. Dividends grow at 6% forever and investors require 13%. What are (a) the price today and (b) the price four years from today?

a. $25.71; $32.46
b. $27.26; $34.41
c. $27.26; $32.46
d. $27.26; $36.48

17. Mirae Logistics paid a dividend of $1.20 four years ago, and the dividend just paid was $1.62. Dividends will keep growing at the same average annual rate. Investors require 14%. What is the price today?

a. $33.56
b. $26.09
c. $28.13

18. Bitgoeul Biotech just paid a dividend of $2.00. Dividends grow 25% next year, 18% the year after and 12% in year 3, then 6% forever. The required return is 14%. What is the price today?

a. $32.61
b. $34.57
c. $26.50
d. $36.24

19. Hanbit Robotics will pay no dividend for the next three years. It will pay $2.00 at the end of year 4 and $2.50 at the end of year 5, after which dividends grow at 4% forever. The required return is 15%. What is the price today?

a. $14.14
b. $21.50
c. $13.69
d. $12.29

20. Gyeongbok Software just paid a dividend of $1.50. Dividends grow at 15% a year for five years and at 5% forever after that. The required return is 16%. Using the two-stage model, what is the price today?

a. $20.37
b. $21.02
c. $14.32
d. $19.13

21. Seorabeol Pharma just paid a dividend of $3.00. Dividends grow at 10% a year for eight years and then at 3% forever. The required return is 11%. What is the price today?

a. $57.93
b. $38.63
c. $23.05
d. $58.97

22. Namsan Electronics just reported EPS of $3.20, and EPS is expected to grow 8% next year. It pays no dividend. Comparable firms trade at forward P/E ratios of 14, 17, 13 and 16. Using the average benchmark, what should the stock sell for?

a. $48.00
b. $51.84
c. $58.75
d. $55.30

23. Cheongdam Mobility had sales of $480 million and a net loss of $36 million. It has 24 million shares and pays no dividend. Comparable firms have price-sales ratios of 2.1, 2.6, 1.9 and 2.4, and an average P/E of 22. Estimate the stock price.

a. −$33.00
b. $48.00
c. $45.00
d. $1,080.00

24. A firm just paid a dividend of $1.00. Dividends grow at 30% for the next two years, at 15% in year 3, and at 5% forever after that. The required return is 13%. What is the price today?

a. $20.66
b. $21.50
c. $19.47
d. $25.51

25. A firm just reported EPS of $2.50, and EPS is expected to grow 10% next year. Comparable firms trade at trailing P/E ratios (price divided by the most recent EPS) of 18, 20, 16 and 22. Using the average benchmark, what should the stock sell for?

a. $52.25
b. $47.50
c. $45.00
d. $55.00

26. Why is a share of stock harder to value than a bond?

a. Stock valuation uses a different principle than bond valuation.
b. Dividends are contractual obligations, but the discount rate is unknown.
c. The market price of a stock can be used to back out its required return exactly.
d. Dividends are discretionary, there is no maturity date, and both R and g must be estimated.

27. A firm pays no dividend now. It is expected never to pay any dividend, and no one who would pay one will ever buy it. What does the dividend discount model imply?

a. The shares are valued at book value.
b. The model cannot be used, so only P/E applies.
c. The shares are worthless.
d. The shares are worth the discounted sales.

28. An investor says: "I am not buying this stock for its dividends, I am buying it to sell it higher." Which response is correct?

a. The dividend discount model does not apply to such investors.

b. The model values only stocks held to maturity.

c. The model already contains this: the future price is itself the present value of later dividends.

d. Capital gains are not part of the model.

29. A board skips the common dividend for several years. What is the legal consequence?

a. The firm is in default.

b. Creditors may demand immediate repayment.

c. The unpaid dividends become a debt of the firm.

d. None: dividends are discretionary, so skipping them cannot cause default or bankruptcy.

30. Which statement distinguishes a dealer from a broker?

a. A broker holds inventory and a dealer earns only a commission.

b. A dealer buys and sells for its own account and bears the risk; a broker matches trades for a commission and takes no position.

c. Both take the other side of the customer's trade.

d. A dealer arranges trades and never holds shares.

Chapter 7: Equity Markets and Stock Valuation — Answers

1. c
Rule: the model works from any date, using the dividend that follows
Step 1: $D_6 = 2.00 \times (1.05)^6 = 2.68$
Step 2: $P_5 = D_6 / (R - g) = 2.68 / 0.07 = \$38.29$
Check: $P_0 = 2.10 / 0.07 = 30.00$ and $P_0 \times (1.05)^5 = 38.29$.

2. c
Step 1: $P_0 = D_1 / (R - g) = 1.00 / (0.09 - 0.10) = -\$100$
Step 2: A negative or infinite price means the assumption is impossible
Rule: the model requires $R > g$
no firm can outgrow its required return forever.

3. c
Step 1: $D_1 = 2.30$
$D_2 = 2.645$
$D_3 = 3.04175$
Step 2: $D_4 = 3.04175 \times 1.05 = 3.1938$
Step 3: $P_3 = D_4 / (R - g) = 3.1938 / 0.07 = 45.63$
Step 4: PV of dividends = 2.0536 + 2.1086 + 2.1651 = 6.33
Step 5: PV of $P_3 = 45.63 / (1.12)^3 = 32.48$
Step 6: $P_0 = 6.33 + 32.48 = \$38.80$.

4. c
Step 1: Apply the model one period before the first dividend, at date 3
Step 2: $P_3 = D_4 / (R - g) = 1.20 / (0.13 - 0.05) = \$15.00$
Step 3: PV $= 15.00 / (1.13)^3 = \$10.40$
Watch for: the model gives the price at date 3, so discount 3 periods, not 4.

5. d
Zero growth: $D_1 = D_0$ here, and only here
Constant growth: use D_1 in the numerator and require $R > g$
Nonconstant growth: discount P_t by t, not t + 1
Multiples: everything depends on the word "comparable".

6. d
Legal form: preferred stock is equity
Tax: the dividend is paid from after-tax income and is not deductible
Default: unpaid dividends are not a debt, so skipping one cannot cause default
Ranking: preferred holders rank behind every creditor.

7. d
Step 1: $D_1 = 2.00 \times 1.08 = 2.16$
Stage 1 PV $= D_1 / (R - g_1) \times [1 - ((1 + g_1) / (1 + R))^5]$
Stage 1 PV $= 2.16 / 0.02 \times [1 - (1.08 / 1.10)^5] = 9.47$
Step 2: $D_6 = 2.00 \times (1.08)^5 \times 1.04 = 3.06$
Step 3: $P_5 = D_6 / (R - g_2) = 3.06 / 0.06 = 50.94$
Step 4: PV of $P_5 = 50.94 / (1.10)^5 = 31.63$
Step 5: $P_0 = 9.47 + 31.63 = \$41.10$.

8. c
Setup: $P_0 = D_0 \times (1 + g) / (R - g)$
Step 1: $50 = 2.00 \times (1 + g) / (0.11 - g)$
Step 2: $5.50 - 50g = 2.00 + 2.00g$
Step 3: $52g = 3.50$, so $g = 6.73\%$
Check: $D_1 = 2.1346$ and $2.1346 / (0.11 - 0.0673) = 50.00$.

9. b
Formula: $R = D_1 / P_0 + g$
Step 1: $D_1 = 1.80 \times 1.05 = 1.89$
Step 2: Dividend yield $= 1.89 / 36 = 5.25\%$
Step 3: $R = 5.25\% + 5.00\% = 10.25\%$
Watch for: the dividend just paid is D_0, so grow it first.

10. c
Step 1: Retention ratio $b = 1 - 0.40 = 0.60$
Step 2: $g = ROE \times b = 0.15 \times 0.60 = 9\%$
Step 3: $D_1 = 0.40 \times 3.00 = 1.20$
Step 4: $P_0 = D_1 / (R - g) = 1.20 / (0.13 - 0.09) = \$30.00$
Idea: growth comes from earnings the firm retains and reinvests at its ROE.

11. c
Step 1: At R = 10%: $P_0 = 2.00 / (0.10 - 0.05) = \$40.00$
Step 2: At R = 11%: $P_0 = 2.00 / (0.11 - 0.05) = \$33.33$
Step 3: Change $= 33.33 / 40.00 - 1 = -16.7\%$
Lesson: the price is highly sensitive to R and g because R − g is a small number.

12. c
Step 1: Dividend yield $= R - g = 12\% - 8\% = 4\%$
Step 2: $D_1 = 0.04 \times 60 = \$2.40$
Step 3: $D_0 = D_1 / (1 + g) = 2.40 / 1.08 = \$2.22$
Watch for: \$2.40 is next year's dividend, not the one just paid.

13. d
Step 1: $P_0 = D_1 / (R - g)$ and $D_1 = \text{Payout} \times E_1$
Step 2: $P_0 / E_1 = \text{Payout} / (R - g)$
Step 3: $P/E = 0.50 / (0.12 - 0.06) = 8.33$
Reading: higher growth or a lower required return raises the justified P/E.

14. b
Step 1: Quarterly dividend $D = 0.075 \times 100 / 4 = \$1.875$
Step 2: Quarterly rate $R = (1 + 0.09/12)^3 - 1 = 2.2669\%$
Step 3: $P_0 = D / R = 1.875 / 0.022669 = \$82.71$
Watch for: $0.09 / 4 = 2.25\%$ ignores monthly compounding and gives \$83.33.

15. d
Step 1: $P = D / R$ gives the value one period before the first payment, at date 2
Step 2: $P_2 = 4.50 / 0.12 = \$37.50$
Step 3: $P_0 = 37.50 / (1.12)^2 = \$29.89$
Watch for: discounting \$37.50 by three years gives \$26.69.

16. b
Step 1: $P_0 = D_0(1 + g) / (R - g) = 1.80 \times 1.06 / 0.07 = \$27.26$
Step 2: $P_4 = D_5 / (R - g) = 1.80 \times (1.06)^5 / 0.07 = \$34.41$
Check: $27.26 \times (1.06)^4 = 34.41$, because the price grows at g.

17. c
Step 1: $1.20 \times (1 + g)^4 = 1.62$, so $g = (1.35)^{1/4} - 1 = 7.79\%$
Step 2: $D_1 = 1.62 \times 1.0779 = 1.7462$
Step 3: $P_0 = 1.7462 / (0.14 - 0.0779) = \$28.13$
Watch for: 35% / 4 = 8.75% is a simple average and overstates g.

18. d
Step 1: $D_1 = 2.50$
$D_2 = 2.95$
$D_3 = 3.304$
Step 2: $P_3 = D_4 / (R - g) = 3.304 \times 1.06 / 0.08 = 43.778$
Step 3: $P_0 = 2.50 / 1.14 + 2.95 / (1.14)^2 + (3.304 + 43.778) / (1.14)^3 = \$36.24$
Watch for: P_3 uses D_4 and is discounted 3 years, not 4.

19. a
Step 1: $P_5 = D_6 / (R - g) = 2.50 \times 1.04 / 0.11 = 23.636$
Step 2: $P_0 = 2.00 / (1.15)^4 + (2.50 + 23.636) / (1.15)^5 = \$14.14$
Watch for: the three zero years must be counted (CF: F01 = 3), or every later cash flow is discounted too few periods.

20. b
Step 1: First stage = $[1.50 \times 1.15 / (0.16 - 0.15)] \times [1 - (1.15 / 1.16)^5] = 7.308$
Step 2: $D_5 = 1.50 \times (1.15)^5 = 3.0170$
Step 3: $P_5 = 3.0170 \times 1.05 / 0.11 = 28.799$
Step 4: $P_0 = 7.308 + 28.799 / (1.16)^5 = 7.308 + 13.712 = \$21.02$
Watch for: $R - g_1 = 1\%$ is tiny, so keep full precision.

21. d
Step 1: First stage = $[3.00 \times 1.10 / (0.11 - 0.10)] \times [1 - (1.10 / 1.11)^8] = 23.047$
Step 2: $D_8 = 3.00 \times (1.10)^8 = 6.4308$
Step 3: $P_8 = D_9 / (R - g_2) = 6.4308 \times 1.03 / 0.08 = 82.796$
Step 4: $P_0 = 23.047 + 82.796 / (1.11)^8 = 23.047 + 35.927 = \$58.97$
Watch for: P_8 is built from D_9, not D_8.

22. b
Step 1: Benchmark P/E = (14 + 17 + 13 + 16) / 4 = 15
Step 2: $EPS_1 = 3.20 \times 1.08 = 3.456$
Step 3: $P_0 = 15 \times 3.456 = \$51.84$
Watch for: a forward P/E must be applied to next year's EPS
$15 \times 3.20 = \$48.00$ mismatches them.

23. c
Step 1: Sales per share = 480 / 24 = \$20.00
Step 2: Benchmark P/S = (2.1 + 2.6 + 1.9 + 2.4) / 4 = 2.25
Step 3: $P_0 = 2.25 \times 20.00 = \$45.00$
Why not P/E: EPS = −36 / 24 = −\$1.50, so $22 \times (-1.50)$ gives a meaningless negative price.

24. b
Step 1: $D_1 = 1.30$
$D_2 = 1.69$
$D_3 = 1.9435$
Step 2: $P_3 = D_4 / (R - g) = 1.9435 \times 1.05 / 0.08 = 25.508$
Step 3: $P_0 = 1.30 / 1.13 + 1.69 / (1.13)^2 + (1.9435 + 25.508) / (1.13)^3 = \$21.50$
Watch for: P_3 uses D_4 and is discounted 3 years, not 4.

25. b
Step 1: Benchmark P/E = (18 + 20 + 16 + 22) / 4 = 19
Step 2: A trailing P/E is applied to the most recent EPS: \$2.50
Step 3: $P_0 = 19 \times 2.50 = \$47.50$
Watch for: applying a trailing P/E to next year's EPS ($19 \times 2.75 = \$52.25$) mismatches the ratio and the earnings.

26. d
Same principle: value = PV of expected cash flows
Problem 1: a dividend is declared at the board's discretion and can be cut to zero without default
Problem 2: a share has no maturity, so the stream is infinite with no face value to anchor it
Problem 3: both the required return and the growth rate must be estimated, and the price is very sensitive to both.

27. c
Principle: price = PV of every dividend the share will ever pay
Firm that eventually pays: the dividend may be far off, but it must be there
Firm that never pays and is never bought by anyone who would: no cash flow to discount
Answer: the shares are worthless.

28. c
Step 1: One-period holding value $P_0 = (D_1 + P_1) / (1 + R)$
Step 2: $P_1 = (D_2 + P_2) / (1 + R)$
Step 3: Substitute again and again
the price terms disappear
Result: only dividends remain
The objection is contained within the model, not a refutation of it.

29. d
Dividends are declared at the board's discretion
Unpaid dividends are not a debt of the firm
Result: failing to pay them cannot cause default or bankruptcy
Contrast: a missed coupon is a default.

30. b
Dealer: trades for its own account, holds an inventory, takes the other side of your trade and bears the risk
Broker: arranges a trade between two other parties, never takes a position
Pay: a broker earns a commission for matching, not for bearing risk.

Business Finance Midterm

Written Calculations · Chapters 2–7 · 15 questions

Exhibit A · Harborview Outfitters, Inc.

Used in questions 1–3

Balance sheets, December 31 ($ millions)

Item	2024	2025
Assets		
Cash	150	200
Accounts receivable	600	700
Inventory	850	900
Total current assets	1,600	1,800
Net fixed assets	4,400	4,600
Total assets	6,000	6,400
Liabilities and owners' equity		
Accounts payable	520	560
Notes payable	280	240
Total current liabilities	800	800
Long-term debt	1,800	1,900
Common stock and paid-in surplus	1,400	1,500
Retained earnings	2,000	2,200
Total owners' equity	3,400	3,700
Total liabilities and owners' equity	6,000	6,400

2025 income statement ($ millions)

Item	2025
Net sales	4,800
Cost of goods sold	2,900
Depreciation	500
Earnings before interest and taxes	1,400
Interest paid	200
Taxable income	1,200
Taxes (21%)	252
Net income	948
Dividends	748
Addition to retained earnings	200

1. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Compute the 2025 cash flow from assets in two ways: (a) from the asset side, CFFA = OCF − NCS − ΔNWC; (b) from the financing side, CFFA = CF to creditors + CF to stockholders, where net new borrowing is the change in long-term debt and net new equity is the change in common stock and paid-in surplus. Write each formula and show that both give the same figure.

2. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Using 2025 year-end figures, compute the profit margin, total asset turnover and equity multiplier, combine them into ROE with the DuPont identity, and confirm the result with NI / TE.

3. [Exhibit A] Harborview Outfitters, Inc. ($ millions). For 2025, compute the current ratio, the quick ratio, the total debt ratio and the debt-equity ratio. Write each formula.

4. Assume the tax schedule in the table. A taxpayer reports taxable income of $200,000. Compute the total tax, the average tax rate and the marginal tax rate.

Tax schedule

Taxable income	Rate
Up to $50,000	10%
$50,001 – $150,000	20%
Above $150,000	30%

5. An investment of $12,000 today will be worth $20,000 in 7 years. (a) What annual rate does it earn? (b) At that rate, how many years would it take $12,000 to grow to $30,000? Write the equation for each part before solving.

6. You deposit $3,000 today and another $5,000 in 18 months. The account pays 6%, compounded monthly. How much is in the account at the end of 4 years? Show the value of each deposit separately.

7. You borrow $12,000 for 3 years at 7.2%, compounded monthly, repaid in equal monthly payments. Compute (a) the monthly payment, (b) the interest in the first payment and (c) the loan balance after the 12th payment.

8. A two-year office lease requires $1,200 at the beginning of every month. The relevant rate is 7.2%, compounded monthly. Value it first as an ordinary annuity and then as the annuity due it actually is.

9. A $22,000 car can be financed with 48 monthly payments of $540. Write the equation for the monthly rate, then report the monthly rate, the quoted annual rate (APR) and the effective annual rate.

10. Lender A charges 11.8%, compounded monthly. Lender B charges 12%, compounded semiannually. Convert both quotes to effective annual rates and state which loan is cheaper.

11. A $1,000 bond pays a 6% coupon semiannually and matures in 12 years. Price it when comparable bonds yield 7% and when they yield 5% (quoted annual rates), and say whether it sells at a premium or a discount in each case.

12. A $1,000 bond pays a 5% coupon semiannually, matures in 8 years and trades at $940. Write the pricing equation, then find the six-month yield, the quoted YTM, the effective annual yield and the current yield, and rank the coupon rate, current yield and YTM.

13. A company just paid a dividend of $1.80 per share. Dividends grow at 4% a year forever and investors require 12%. Compute (a) today's price, (b) the expected price three years from today and (c) the dividend yield and capital gains yield.

14. A firm just paid a $1.50 dividend. Dividends will grow at 15% for the next two years and at 4% forever after that. The required return is 12%. Write out each dividend, the terminal price and the value today.

15. A firm will pay no dividend for the next three years. It will pay $1.00 at the end of year 4, and that dividend will then grow at 5% forever. The required return is 15%. Value the stock today, showing the date at which you apply the growth model.

Business Finance Midterm — Model answers

1. Ch.2 · Cash flow from assets, two ways
OCF = EBIT + depreciation − taxes = 1,400 + 500 − 252 = 1,648
NCS = ending net FA − beginning net FA + depreciation = 4,600 − 4,400 + 500 = 700
ΔNWC = (1,800 − 800) − (1,600 − 800) = 1,000 − 800 = 200
CFFA = OCF − NCS − ΔNWC = 1,648 − 700 − 200 = 748
CF to creditors = interest paid − net new borrowing = 200 − (1,900 − 1,800) = 100
CF to stockholders = dividends paid − net new equity = 748 − (1,500 − 1,400) = 648
Check = 100 + 648 = 748, the same as the asset-side figure
Key results: 1,648 · 700 · 748 · 100 · 648

2. Ch.3 · DuPont identity
PM = NI / sales = 948 / 4,800 = 19.75%
TAT = sales / TA = 4,800 / 6,400 = 0.75
EM = TA / TE = 6,400 / 3,700 = 1.73
ROE = PM × TAT × EM = 0.1975 × 0.75 × 1.73 = 25.62%
Check = NI / TE = 948 / 3,700 = 25.62%
Key results: 19.75 · 0.75 · 1.73 · 25.62

3. Ch.3 · Liquidity and leverage
Current ratio = CA / CL = 1,800 / 800 = 2.25
Quick ratio = (CA − inventory) / CL = (1,800 − 900) / 800 = 1.13
Total debt = TA − TE = 6,400 − 3,700 = 2,700
Total debt ratio = TD / TA = 2,700 / 6,400 = 0.42
D/E = TD / TE = 2,700 / 3,700 = 0.73
Key results: 2.25 · 1.13 · 0.42 · 0.73

4. Ch.2 · Average and marginal tax rates
Tax = 0.10 × 50,000 + 0.20 × 100,000 + 0.30 × 50,000 = 5,000 + 20,000 + 15,000 = 40,000
Average rate = total tax / taxable income = 40,000 / 200,000 = 20%
Marginal rate = rate on the next dollar of income = 30%
Key results: 40,000 · 20% · 30%

5. Ch.4 · Rate and time
(a) Equation: $20{,}000 = 12{,}000 \times (1 + r)^{7}$
r = $(20{,}000 / 12{,}000)^{1/7} - 1 = 7.57\%$
BA II Plus: 7 N, −12000 PV, 0 PMT, 20000 FV, CPT I/Y = 7.57
(b) Equation: $30{,}000 = 12{,}000 \times (1.0757)^{t}$
t = ln(2.5) / ln(1.0757) = 12.56 years
BA II Plus: 7.57 I/Y, −12000 PV, 0 PMT, 30000 FV, CPT N = 12.56
Key results: 7.57 · 12.56

6. Ch.5 · Two deposits
r = 6% / 12 = 0.5% per month
FV = $3{,}000 \times (1.005)^{48} + 5{,}000 \times (1.005)^{30}$
First deposit = $3{,}000 \times (1.005)^{48} = 3{,}811.47$ (48 months)
Second deposit = $5{,}000 \times (1.005)^{30} = 5{,}807.00$ (30 months)
Total = 9,618.47
BA II Plus: 48 N, 0.5 I/Y, 0 PMT, −3000 PV, CPT FV = 3,811.47
30 N, −5000 PV, CPT FV = 5,807.00
Key results: 3,811.47 · 5,807.00 · 9,618.47

7. Ch.5 · Amortizing loan
r = 7.2% / 12 = 0.6% per month, N = 36
Payment: $12{,}000 = C \times [1 - 1/(1.006)^{36}] / 0.006$, so C = 371.62
First interest = 0.006 × 12,000 = 72.00
Balance after 12 = PV of the 24 remaining payments = $371.62 \times [1 - 1/(1.006)^{24}] / 0.006$ = 8,283.46
BA II Plus: 36 N, 0.6 I/Y, 12000 PV, 0 FV, CPT PMT = −371.62
AMORT: P1 = 1, P2 = 12, BAL = 8,283.46
Key results: 371.62 · 72.00 · 8,283.46

8. Ch.5 · Lease value (annuity due)
r = 0.6% per month, N = 24
Ordinary annuity = $1{,}200 \times [1 - 1/(1.006)^{24}] / 0.006$ = 26,747.92
Annuity due = ordinary annuity × (1 + r) = 26,747.92 × 1.006 = 26,908.41
BA II Plus: 24 N, 0.6 I/Y, −1200 PMT, 0 FV, CPT PV = 26,747.92, then × 1.006 (or BGN mode, switched off afterwards)
Key results: 26,747.92 · 26,908.41

9. Ch.5 · Rate on a car loan
Formula: $22{,}000 = 540 \times [1 - 1/(1 + r)^{48}] / r$, solved by iteration
Monthly rate = 0.6902%
APR = 0.6902% × 12 = 8.28%
EAR = $(1.006902)^{12} - 1$ = 8.60%
BA II Plus: 48 N, 22000 PV, −540 PMT, 0 FV, CPT I/Y = 0.6902
ICONV: NOM = 8.28, C/Y = 12, CPT EFF = 8.60
Key results: 0.69 · 8.28 · 8.60

10. Ch.5 · Comparing lenders
EAR(A) = $(1 + 0.118/12)^{12} - 1$ = 12.46%
EAR(B) = $(1 + 0.12/2)^{2} - 1$ = 12.36%
Decision: borrow from Lender B, which has the lower EAR
ICONV: NOM = 11.8, C/Y = 12, CPT EFF = 12.46
NOM = 12, C/Y = 2, CPT EFF = 12.36
Key results: 12.46 · 12.36 · Lender B

11. Ch.6 · Bond pricing
Inputs: N = 12 × 2 = 24, PMT = 60 / 2 = 30, FV = 1,000
At 7%: $P = 30 \times [1 - 1/(1.035)^{24}] / 0.035 + 1{,}000 / (1.035)^{24}$ = 919.71, a discount (coupon rate < yield)
At 5%: $P = 30 \times [1 - 1/(1.025)^{24}] / 0.025 + 1{,}000 / (1.025)^{24}$ = 1,089.42, a premium (coupon rate > yield)
BA II Plus: 24 N, 3.5 I/Y, 30 PMT, 1000 FV, CPT PV = −919.71
2.5 I/Y, CPT PV = −1,089.42
Key results: 919.71 · 1,089.42 · discount · premium

12. Ch.6 · Yield to maturity
Formula: $940 = 25 \times [1 - 1/(1 + r)^{16}] / r + 1{,}000 / (1 + r)^{16}$, solved by iteration
Six-month yield = 2.98%
Quoted YTM = 2.98% × 2 = 5.95%
Effective annual yield = $(1.0298)^{2} - 1$ = 6.04%
Current yield = annual coupon / price = 50 / 940 = 5.32%
Ranking: coupon rate 5.00% < current yield 5.32% < YTM 5.95% (a discount bond)
BA II Plus: 16 N, −940 PV, 25 PMT, 1000 FV, CPT I/Y = 2.98
Key results: 2.98 · 5.95 · 6.04 · 5.32

13. Ch.7 · Constant growth
$D_1 = D_0 \times (1 + g) = 1.80 \times 1.04 = 1.872$
$P_0 = D_1 / (R - g) = 1.872 / 0.08 = 23.40$
$P_3 = D_4 / (R - g) = 1.80 \times (1.04)^4 / 0.08 = 2.1057 / 0.08 = 26.32$
Dividend yield = $D_1 / P_0 = 1.872 / 23.40 = 8\%$
Capital gains yield = g = 4% (R = 8% + 4% = 12%)
Key results: 23.40 · 26.32 · 8% · 4%

14. Ch.7 · Nonconstant growth
$D_1 = 1.50 \times 1.15 = 1.725$
$D_2 = 1.725 \times 1.15 = 1.98375$
$D_3 = 1.98375 \times 1.04 = 2.0631$
$P_2 = D_3 / (R - g) = 2.0631 / (0.12 - 0.04) = 25.79$
$P_0 = 1.725 / 1.12 + (1.98375 + 25.79) / (1.12)^2 = 1.540 + 22.140 = 23.68$
BA II Plus CF: CF0 = 0, C01 = 1.725, C02 = 27.773, NPV I = 12, CPT = 23.68
Key results: 1.725 · 1.98375 · 2.0631 · 25.79 · 23.68

15. Ch.7 · A firm that pays nothing yet
Growth model at date 3, one period before the first dividend: $P_3 = D_4 / (R - g) = 1.00 / (0.15 - 0.05) = 10.00$
$P_0 = P_3 / (1.15)^3 = 10.00 / 1.5209 = 6.58$
BA II Plus: 3 N, 15 I/Y, 0 PMT, 10 FV, CPT PV = −6.58
Key results: 10.00 · 6.58